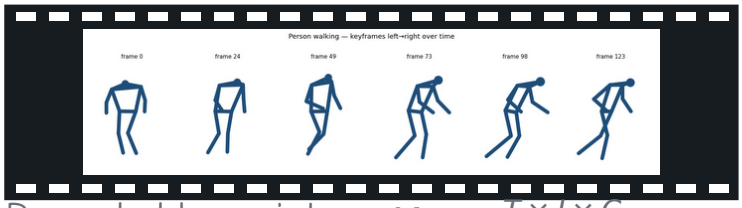
1 Pose source + participant split



Recorded keypoints $X \in \mathbb{R}^{T \times J \times C}$

train val. test

Transform statistics from training only

2 Map joints → deterministic measurements

Range

max - min

Asymmetry

Left Right

Stability

trunk / pelvis

Smoothness

mean $|\Delta^3 p|$

Keep supported joint roles; exclude capture-quality terms in the shared suite.

3 Render one input view

Numerical feature values

$z = (f - \mu_{\text{train}})/\sigma_{\text{train}}$

Fitted once on training records

Deterministic summary

Numbers + fixed descriptors

No answer label / identity fields

4 Evidence + candidate classes

same features

[selected evidence view]

A B C D

Zero-shot controls

$\varnothing$ $e_i \longrightarrow e_{\pi(i)}$

Candidate-order controls

D C B A reverse / shuffle

Qwen2.5-7B (ZS)

Frozen; temperature 0

Summary LoRA-SFT

r = 16 / 3 epochs

LR / SVM

Signal probe

Parse the last complete candidate mention

No match → invalid → error (never dropped)

Accuracy / macro-F1 / κ · per-class errors · participant checks